Multimodal Visual Understanding（Voice Version）

Before running the function, you need to close the App and large programs. For the closing method, refer to [4. Preparation] - [1. Manage APP control services].

1. Function Description

After the program runs, wake up the voice module and ask questions about vision or describe the current scene. The program will capture an image of the current environment and upload it to the Alibaba Cloud platform, then the vision model will analyze the image with the question and finally provide an answer.

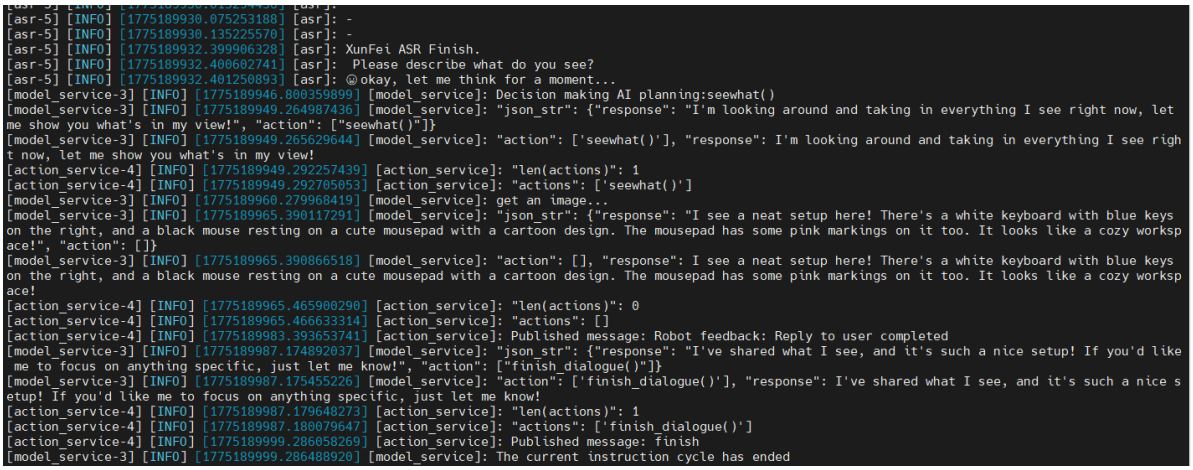
2. Startup

After entering the Docker container, enter the following commands in the Docker terminal.

```
ros2 launch largemodel largemodel_control.launch.py
```

After waking up the module, give vision-related commands. You can refer to the following example,

```
Please describe what do you see?
```



```
[asr-5] [INFO] [1775189930.075253188] [asr]: -
[asr-5] [INFO] [1775189930.135225570] [asr]: -
[asr-5] [INFO] [1775189932.399906328] [asr]: XunFei ASR Finish.
[asr-5] [INFO] [1775189932.400602741] [asr]:  Please describe what do you see?
[asr-5] [INFO] [1775189932.401250893] [asr]: 😀okay, let me think for a moment...
[model_service-3] [INFO] [1775189946.800359899] [model_service]: Decision making AI planning:seewhat()
[model_service-3] [INFO] [1775189949.264987436] [model_service]: "json_str": {"response": "I'm looking around and taking in everything I see right now, let
me show you what's in my view!", "action": ["seewhat()"]}
[model_service-3] [INFO] [1775189949.265629644] [model_service]: "action": ['seewhat()'], "response": I'm looking around and taking in everything I see righ
t now, let me show you what's in my view!
[action_service-4] [INFO] [1775189949.292257439] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775189949.292705053] [action_service]: "actions": ['seewhat()']
[model_service-3] [INFO] [1775189960.279968419] [model_service]: get an image...
[model_service-3] [INFO] [1775189965.390117291] [model_service]: "json_str": {"response": "I see a neat setup here! There's a white keyboard with blue keys
on the right, and a black mouse resting on a cute mousepad with a cartoon design. The mousepad has some pink markings on it too. It looks like a cozy worksp
ace!", "action": []}
[model_service-3] [INFO] [1775189965.390866518] [model_service]: "action": [], "response": I see a neat setup here! There's a white keyboard with blue keys
on the right, and a black mouse resting on a cute mousepad with a cartoon design. The mousepad has some pink markings on it too. It looks like a cozy worksp
ace!
[action_service-4] [INFO] [1775189965.465900290] [action_service]: "len(actions)": 0
[action_service-4] [INFO] [1775189965.466633314] [action_service]: "actions": []
[action_service-4] [INFO] [1775189983.393653741] [action_service]: Published message: Robot feedback: Reply to user completed
[model_service-3] [INFO] [1775189987.174892037] [model_service]: "json_str": {"response": "I've shared what I see, and it's such a nice setup! If you'd like
 me to focus on anything specific, just let me know!", "action": ["finish_dialogue()"]}
[model_service-3] [INFO] [1775189987.175455226] [model_service]: "action": ['finish_dialogue()'], "response": I've shared what I see, and it's such a nice s
etup! If you'd like me to focus on anything specific, just let me know!
[action_service-4] [INFO] [1775189987.179648273] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775189987.180079647] [action_service]: "actions": ['finish_dialogue()']
[action_service-4] [INFO] [1775189999.286058269] [action_service]: Published message: finish
[model_service-3] [INFO] [1775189999.286488920] [model_service]: The current instruction cycle has ended
```

The captured photo is saved at:
`LargeModel_ws/install/largemodel/share/largemodel/resources_file/image.png`

3. Core Code Analysis seehat

You can refer to the content in **3. Core Code Analysis seehat** from tutorial [17. AI Model - Text Version] - [2. Multimodal Visual Understand]. The voice version and text version have the same action functions, only the task command input method is different.